

Empirical welfare analysis: when preferences matter

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Abstract The conditional equality and egalitarian equivalence criteria were proposed by Fleurbaey (Fairness, responsibility, and welfare, Oxford University Press, Oxford, 2008) to provide better foundations to interpersonal comparisons in the context of heterogeneous preferences and multidimensional welfare. The first implementations of the egalitarian equivalence criterion follow an approach where the preferences are captured at the group level (based on socio-demographic variables) rather than at the individual level. Our contribution is to extend these models by using information on individual preferences, derived from the potential discrepancy between the group level optimal choice and the revealed choice of the individuals. We implement and compare the conditional equality and egalitarian equivalence criteria on a 2004 US microeconomic dataset and find that these criteria are relatively consistent in the identification of the worst-off. We also show that up to 18 % of the worst-off are no longer categorized as worst-off when the empirical approach accounts for individual preferences.

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1 Introduction

Measuring the welfare of heterogeneous agents and considering the multi-dimensional character of welfare remains a highly difficult exercise. Obtaining a social ordering that fully respects heterogeneous preferences in an ordinal setting requires some restrictions. One contribution of the theory of fair social orderings (developed in [Fleurbaey 2008](#); [Fleurbaey and Maniquet 2005, 2006, 2011](#); [Bossert and Fleurbaey 1996](#)) has been to define restrictive fairness criteria allowing such interpersonal comparisons.

Designing a fair redistribution policy with heterogeneous agents is also a difficult task. It requires to draw a line between fair and unfair inequalities. The latter arises from factors beyond the control of individuals such as innate inabilities or parental education (more generally called circumstances) and the former from factors within the control of individuals, and based on their effort or preferences.

From this typology between responsibility and non-responsibility factors, we can derive two principles (detailed in [Fleurbaey 2008](#)). The first one is the compensation principle, which states that the effect of circumstances (social background, learning abilities, skills) should be neutralized in order to level the playing field or, in other words, give to all the same opportunity set: “equal well-being for equal responsibility”. The second principle is the responsibility principle, usually captured by the liberal reward or neutrality principle, which states that there should be no differentiated intervention when two individuals have the same circumstances, that interventions should not affect/distort responsibility factors: “equal treatment for equal skills”.

These two compensation and neutrality principles cannot be both fully satisfied in general. As [Schokkaert et al. \(2004\)](#) put it: “problems arise if we want to hold individuals fully accountable for differences in outcomes that result from differences in (pure) preferences (e.g., for work effort), while we want to compensate them fully for skill (or ability) differences.” This incompatibility has been proven by [Fleurbaey \(1995\)](#) and [Bossert \(1995\)](#).

A solution proposed by [Fleurbaey \(2008\)](#) consists in setting constraints such that one principle is preferred to the other. The conditional equality and egalitarian equivalence criteria give pre-eminence respectively to the neutrality and compensation principles.

The conditional equality criterion completely fulfills the neutrality principle and partially achieves the compensation for circumstances. The identification of the worst-off is achieved by determining what level of well-being an individual would achieve with some reference responsibility characteristics and his current circumstances. The conditional equality thus remains fully neutral toward individual responsibility factors. We will implement the conditional equality criterion in our empirical illustration with three successive reference responsibility characteristics (from low to high preference for leisure).

The egalitarian equivalence criterion on the contrary completely fulfills the compensation principle, and partially the neutrality principle. The identification of the worst-off is achieved by determining what amount of resources an individual would need, with a reference circumstance and his current responsibility characteristics in order to maintain his current level of well-being. Given a reference circum-

stance, the egalitarian equivalence criterion fully compensates (through a transfer) for unfair inequalities, but can partially be non-neutral to preferences. We will follow [Bargain et al. \(2013\)](#) and [Decoster and Haan \(2015\)](#) in our empirical exercise by taking the wage as a proxy for circumstances and using three reference circumstances: the zero-wage, the minimum wage and the wage egalitarian equivalence criteria.

We apply and compare these criteria on a 2004 US database in view to identify the worst-off, defined as the 10 % individuals most legitimate to benefit from a redistributive transfer. We consider income and leisure as the two sources of well-being, resulting in the traditional consumption-leisure trade-off in labor supply modelling. Accounting for the heterogeneity of individuals requires identifying preferences. The first empirical papers to implement responsibility-sensitive theories of justice are [Almas et al. \(2011\)](#) and [Devooght \(2005\)](#), who used the definition of conditional equality and egalitarian equivalence criteria to measure unfair inequalities for income.

We follow an alternative approach and estimate preferences by means of a static structural discrete choice model of labor supply, similar to [Aaberge et al. \(1995\)](#) and [van Soest \(1995\)](#) and follow the implementation of the egalitarian equivalence criterion detailed in [Decoster and Haan \(2015\)](#) and [Bargain et al. \(2013\)](#). One assumption in their approach is that individual preferences are better captured at the group level (based on socio-demographic variables) rather than at the individual level. This is mainly justified by the practical difficulty to identify individual preferences but also by the idea that individual choices are noisy (not well-informed, errors of judgement) and need to be averaged out. What we propose is a way to partially identify the individual preferences, but also an approach where individual choices are better respected.

Our contribution is to develop a new empirical approach by using information on individual preferences, derived from the potential discrepancy between the group level optimal choice and the revealed choice of the individuals. We show in our sample that 57 % of the individuals made another choice than the one predicted by the model based on socio-demographic group variables. We use these discrepancies to refine our identification of the worst-off according to the egalitarian equivalence criterion. We compare conditional equality and egalitarian equivalence criteria (with and without identification of the individual preferences) and show the differences and similarities. To illustrate the importance of considering individual preferences, we also show that up to 18 % of the worst-off drop from the worst-off category when our method to identify individual preferences is implemented. More generally, our approach allows to refine the evaluation of redistributive policies by measuring to which extent socio-demographic criteria are a good guide to identify the worst-off.

The rest of this paper is organized as follows: Sect. 2 presents the responsibility-sensitive egalitarian principles and the fairness criteria, Sect. 3 explains the empirical approach and Sect. 4 the extension designed to better reflect individual preferences. We finally present the data and results in Sect. 5 and conclude in Sect. 6.

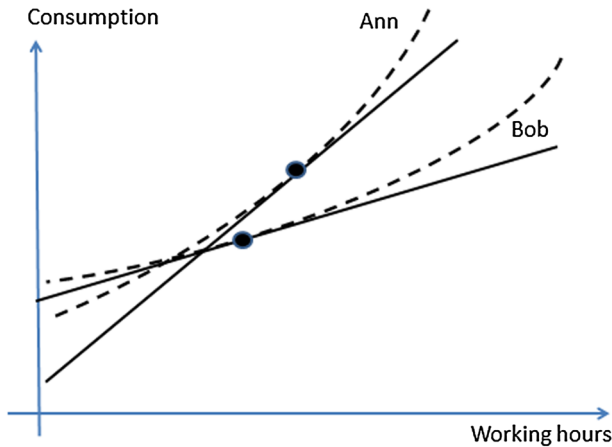


Fig. 1 Choice of Ann and Bob, with different preferences and budget sets

2 Theoretical framework

In this section, we present the utility model, and rely on the terminology and set-up of [Bargain et al. \(2013\)](#) and [Decoster and Haan \(2015\)](#).¹ Individuals have to choose their consumption bundle (c, l) where c stands for consumption (or net income) and l for labor supply and where $(c_i, l_i) \in \mathbb{R}_+ \times [0, 1]$. Their heterogeneous preferences are partially captured through a vector of observed variables z_i . Individual i chooses his bundle (c_i, l_i) by maximizing utility v_i given his budget constraint as follows:

$$(c_i, l_i) = \arg \max [v(c, l; z_i) | (c \leq f(l_i, w_i l))], \quad (1)$$

where $f(\cdot)$ is a function representing the tax benefit system, transforming non-labor income I_i and labor income $w_i l$, with w_i denoting the gross wage specific to individual i , into net income c . In this framework, differences of outcomes between individuals come either from heterogeneous preferences z_i or from the budget constraint, which is determined by both the non-labor income I_i and the gross wage w_i .

Figure 1 illustrates a case in the (c, l) space with the indifference curves (dashed lines) and budget constraints (solid lines) of two individuals, Ann and Bob. Heterogeneous preferences z_i affect the steepness of the indifference curves. Individuals with a lower (higher) preference for leisure will have a flatter (steeper) indifference curve. We see in this example that Bob has a low preference for leisure (flat indifference curve), compared to Ann.

We also illustrate different budget sets in Fig. 1 where Bob has a higher non-labor income and a lower wage than Ann. The same Figure also illustrates the incomparability of agents with heterogeneous preferences. Deciding which of the high consumption high working hours of Ann and the low consumption low working hours bundle of Bob is the best is not straightforward. Additional assumptions are needed to decide

¹ See [Decoster and Haan \(2010\)](#) for the preliminary version of the paper.

whether Bob is better-off or worse-off than Ann. We now formally present the conditional equality and egalitarian equivalence criteria.

The conditional equality criterion fixes a reference responsibility variable, $\bar{z}$, captured in our framework by the steepness of the indifference curve, and computes the maximum well-being that an individual i can reach, $\hat{v}_i$, given his equivalent budget set respecting individual preferences (i.e. a budget with lump-sum tax, $\hat{t}_i$, which would give him his current satisfaction $v(c_i, l_i; z_i)$, as obtained from Eq. 1). The interpersonal comparability is thus realized through the comparison of the well-being $\hat{v}_i$ that the individuals can reach given the reference preference and their respective equivalent budget sets. More formally:

$$\hat{v}_i = \max [\hat{v}(c, l; \bar{z}) | c \leq w_i l - \hat{t}_i] \quad (2)$$

where $\hat{t}_i$ is implicitly defined by

$$v(c_i, l_i; z_i) = \max [v(c, l; \bar{z}) | c \leq w_i l - \hat{t}_i]. \quad (3)$$

The variants of the reference preferences used in the Sect. 3 go from a low preference for leisure (flat indifference curve) to a high preference for leisure (steep indifference curve). The conditional equality criterion is illustrated in Fig. 2, where we only represent Bob for readability. The dashed curve represents the actual/computed indifference curve of Bob and the big dot is the consumption-leisure bundle chosen by Bob given his actual budget set. The solid line is the equivalent budget set of Bob. The lump-sum transfer $\hat{t}_{Bob}$ is set such that his equivalent budget enables him to reach his current indifference curve. The dotted line represents the indifference curve associated with the reference preferences. Given Bob's so defined equivalent budget set, we find that the higher indifference curve that Bob can reach is $\hat{v}_{Bob}$. This indifference curve is to

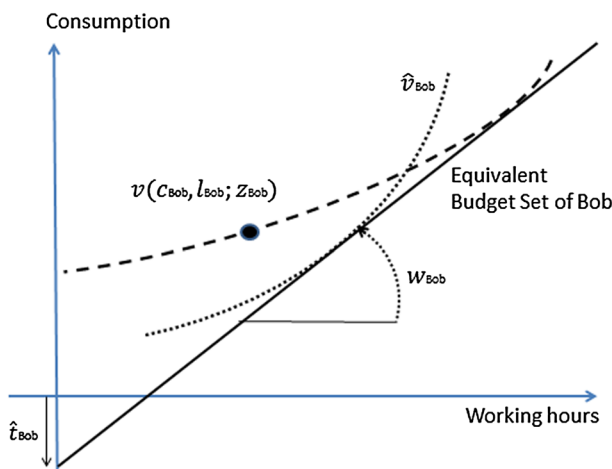


Fig. 2 Conditional equality—given reference preferences and equivalent budget sets, the worst-off is the one who reaches the lowest indifference curve

be compared with the respective indifference curve that Ann can reach given her own equivalent budget set.

Thanks to the equivalent budget set, the conditional equality criterion respects individual preferences. Indeed a better bundle for an individual corresponds to a bigger equivalent budget, and reference preferences always prefer bigger budgets as well (Fleurbaey 2008, p. 129). In addition, the choice of the reference preference, $\hat{v}$, has a potentially large impact on the identification of the worst-off. The lower (higher) the preference for leisure, the more likely the low(high)-wage individuals will be considered as worst-off. In other words, a lower (higher) reference preference for leisure will identify the less (more) productive—those with bad (good) circumstances—as the worst-off.

Egalitarian equivalence is our second criterion. It fixes a reference set of circumstances (captured in our framework by the wage and the non-labor income) and looks at how the reference budget constraint must be shifted or rotated to allow the individual to reach his current well-being, given his individual preferences. Interpersonal comparability of welfare levels is obtained by measuring welfare through nested budget sets B_λ , where the set B_λ is implicitly defined by:

$$u(c_i, l_i; z_i) = \max [u(c, l; z_i) | (c, l) \in B_\lambda]. \quad (4)$$

The chosen bundle (c_i, l_i) on a given indifference curve is evaluated by indexing the curves by means of these reference sets, where $\lambda \leq \lambda'$ if and only if $B_\lambda \subseteq B_{\lambda'}$. The situation of individual i is as good as λ is high. Our implementation relies on three specific versions of B_λ .

The first version defines the reference budget set as:

$$B_{\lambda^{Min}} = \left\{ (c, l) | c \leq \lambda^{Min} + w^{Min}l, \quad l \leq 1 \right\}. \quad (5)$$

In this version, called minimum wage egalitarian equivalence, the reference wage is set at the minimum wage w^{Min} , and the dominant subset is the one where λ^{Min} is the largest. This metric is illustrated in Fig. 3 where we note that Ann is the worst-off, with the lowest transfer required to reach her current well-being (smallest vertical arrow in the bottom left area of the figure).

The second version, called zero-wage egalitarian equivalence, sets the reference wage to zero.

$$B_{\lambda^{Zero}} = \left\{ (c, l) | c \leq \lambda^{Zero}, \quad l \leq 1 \right\}. \quad (6)$$

This metric is illustrated in Fig. 4 where we note that Ann is once again the worst-off, with a transfer required to reach her current well-being smaller than the one of Bob. The choice of the reference wage might have an impact on the identification of the worst-off. Indeed, the smaller the reference wage ($w = 0$), the more likely the individuals with high preference for leisure (Ann) will be considered as worst-off. In other words, a low reference wage metric will favor the “lazy” and be detrimental to the “industrious”.

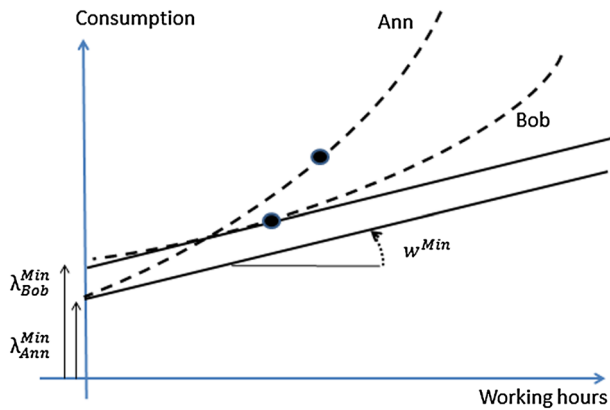


Fig. 3 Minimum wage egalitarian equivalence—given a reference wage set to the minimum wage and given actual preferences, the worst-off is the one who needs the smallest transfer to obtain his current well-being

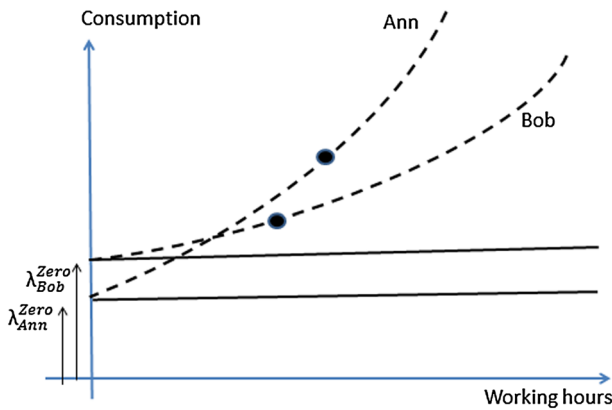


Fig. 4 Zero-wage egalitarian equivalence—even a reference wage set to zero and given actual preferences, the worst-off is the one who needs the smallest transfer to obtain his current well-being

The third version, called wage egalitarian equivalence, sets the non-labor income to zero as follows:

$$B_{\lambda^{Wage}} = \left\{ (c, l) \mid c \leq \lambda^{Wage} l, \quad l \leq 1 \right\}. \quad (7)$$

This metric is illustrated in Fig. 5 where we note that Bob is the worst-off, with a wage required to reach his current well-being lower than the one of Ann.

These metrics are not neutral with respect to preferences since the steepness of individual indifference curves will have an impact on the identification of the worst-off. We present in the next section the empirical procedure followed to estimate the indifference curves.

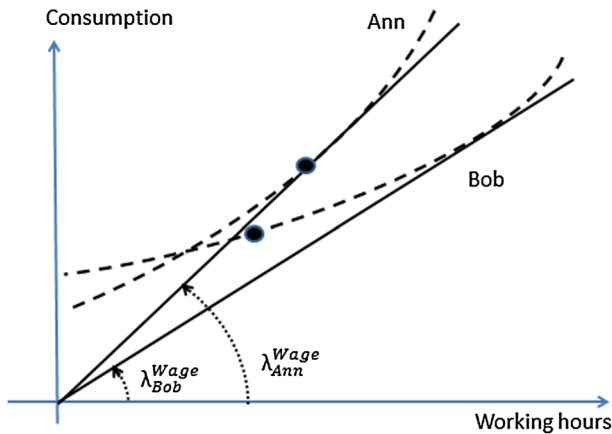


Fig. 5 Wage egalitarian equivalence—given a non-labor income set to zero and given actual preferences, the worst-off is the one who needs the smallest wage to obtain his current well-being

3 Empirical approach

We estimate household preferences by means of a static structural discrete choice model of labor supply, similar to the one developed by [Aaberge et al. \(2000, 2004\)](#), [van Soest \(1995\)](#) and later by [Bargain et al. \(2013\)](#) and [Decoster and Haan \(2015\)](#). The model is static, as it excludes savings. It is discrete to the extent that the model reduces the choice to a finite number of discrete alternatives (working hours).² We restrict households to single adults, with no children, in our application.³

The state specific utility v_{ij} of individual i at discrete state j , where $j = 1, \dots, J$, is composed of a deterministic component $u_{ij}(\cdot)$ and a stochastic component ϵ_{ij} . The stochastic component ϵ_{ij} is not observed and contains the heterogeneity not captured by the variables contained in the vector $\mathbf{z}_i$.

$$v_{ij} = u(c_{ij}, l_{ij}; \mathbf{z}_i) + \epsilon_{ij}. \quad (8)$$

The individual maximizes his utility by choosing his amount of working hours per week from a set of J alternatives, where the alternatives are working 20, 25, $\dots$ 50, or 55 h. The individual is subject to a budget set that depends on his wage rate, the amount of labor time, the non-labor income and the taxes he pays on his gross income. The wage rate is assumed to be constant whatever the amount of labor time picked by the individual. As the actual taxes are non-linear, we use information on income taxes

² This discretization is convenient for the estimation procedure. It also allows to better account for the non-linearities in the budget set (due to progressive taxes). It finally reflects the effective choice of the individuals who have in principle not an infinite number of working hours options. Such choices are usually discrete.

³ This is an alternative to the approach of [Decoster and Haan \(2015\)](#), limited to married households, with the labor choice of married woman as the focus, and the assumption that male income is a non-labor income.

provided by the OECD, estimate the actual individual's budget set and calculate his net income for each of the 8 working hours alternatives.

The budget set can be written as:

$$c_{ij} = f(I_i, w_i l_{ij}), \quad (9)$$

where $f(I_i, w_i l_{ij})$ is the tax function that transforms gross income into net income.

The deterministic part of the utility function is specified as a Cox-Box function (Box and Cox (1964)) as in Aaberge et al. (2000, 2004) and Decoster and Haan (2015):

$$u(c_{ij}, l_{ij}; \mathbf{z}_i) = \beta_c \frac{c_{ij}^{\alpha_c} - 1}{\alpha_c} + \beta_l(\mathbf{z}_i) \frac{(1 - l_{ij})^{\alpha_l} - 1}{\alpha_l}, \quad (10)$$

where β_c , $\beta_l(\cdot)$, α_c and α_l determine the preferences.

This Cox-Box utility specification is convenient as it allows to easily control that the marginal utilities of consumption and leisure are positive (β_c and β_l positive) and decreasing (α_c and α_l smaller than 1). β_c , α_c and α_l are common to all individuals. Heterogeneity among groups of individuals is introduced through $\beta_l(\mathbf{z}_i)$ that depends on socio-demographic variables as follows:

$$\beta_l(\mathbf{z}_i) = \beta_{l_0} + \beta'_{l_1} \mathbf{z}_i, \quad (11)$$

where $\mathbf{z}_i$ includes gender, age, education and ethnic group. The heterogeneity parameters, $\beta_l(\mathbf{z}_i)$, also called taste-shifters, play an important role for the egalitarian equivalence criterion, where the heterogeneity of preferences is fully respected. They also impact the implementation of the conditional equality criterion as the computation of the equivalent budgets depends on the identification of preferences.

The variables that explain the differences in the preference for leisure have been widely used in the literature to explain the determinants for labor supply. They are not really under the individual's control but they are expected to explain differences in preferences for leisure. In fact, Dworkin (1981a,b), Rawls (2006) and Fleurbaey and Maniquet (2005) share the view that we should be neutral with respect to preferences as long as individuals identify themselves with these preferences. We follow the view of these authors. β_l therefore captures individuals' preferences for leisure.

Under the standard assumption that the stochastic component ϵ_{ij} is iid type 1 extreme value distributed, McFadden (1974) showed that the probability for each individual of choosing a given alternative j (Eqs. 12, 13) has a logistic form (Eq. 14) and can be estimated by maximum likelihood as a conditional logit model (Cameron

and Trivedi 2005).⁴

$$Prob(l_i = j) = Prob(v_{ij}(\cdot) > v_{ik}(\cdot)) \quad \forall k \neq j, \quad (12)$$

$$= Prob(\epsilon_{ik} - \epsilon_{ij} < -(u_{ik}(\cdot) - u_{ij}(\cdot))) \quad \forall k \neq j, \quad (13)$$

$$= \frac{\exp u_{ij}(\cdot)}{\sum_{k=1}^J \exp u_{ik}(\cdot)}. \quad (14)$$

We can thus estimate the utility function parameters and compute the state j utility level for each socio-demographic group (based on the vector $\mathbf{z}_i$ of socio-demographic variables). We can then graph the whole indifference curve (more precisely the whole set of discrete points) corresponding to the utility level determined by the (c_{ij}, l_{ij}) combination, by solving c_{ik} for any k working hours choice l_{ik} .

$$(\tilde{c}_{ik}, l_{ik}) = \left(\left[c_{ij}^{\alpha_c} + \frac{\alpha_c}{\beta_c} \left(\epsilon_{ij} - \epsilon_{ik} + \frac{\beta_l(\mathbf{z}_i)}{\alpha_l} ((1 - l_{ij})^{\alpha_l} - (1 - l_{ik})^{\alpha_l}) \right) \right]^{\frac{1}{\alpha_c}}, l_{ik} \right). \quad (15)$$

Since $\epsilon_{ij} - \epsilon_{ik}$ is neither observed nor estimated, we take the expectations and set $\epsilon_{ij} - \epsilon_{ik}$ to zero and Eq. 15 can be accordingly simplified.

In the next Section, we explore how identification of preferences can be refined and go beyond group's preferences to reach part of the individuals preferences.

4 Individual preferences

Decoster and Haan (2015) note that their implementation of the egalitarian equivalence criteria focuses on the deterministic part of the preferences, which “explains only part of the variation in choices for individuals facing the same constraints”. In other words, they “assume that two individuals with the same vector $\mathbf{z}$ do have the same preferences”. This assumption is mainly due to the difficulty of identifying individual preferences but also to the idea that individual choices might be noisy (as a consequence of incomplete information or imperfect processing of the information). Averaging the actual preference across some socio-demographic groups is a way to filter out the noise.

We do not make such assumption. First, we show (below) that individual preferences, not summarized by socio-demographic variables, can be retrieved to some extent in the stochastic component of the utility function. It is therefore possible to go beyond group preferences. Second, we consider that labelling as noise all the information not captured by the socio-demographic variables is excessive, especially when the model does not include an extensive list of socio-demographic variables. We thus provide an alternative to the group preference approach where we take individuals decision as reflecting respectable preference (and not exclusively noise).

⁴ A type 1 extreme value distributed random variable ϵ has density $f(\epsilon) = e^{-\epsilon} \exp(-e^{-\epsilon})$ and cumulative distribution function $F(\epsilon) = \exp(-e^{-\epsilon})$. See Cameron and Trivedi (2005), p. 477 for more details.

Our econometric strategy extends the approach of [McFadden \(1974\)](#) to account for a part of the unobserved heterogeneity. The idea is close to the one developed by [Bonin and Schneider \(2006\)](#). Although their focus relates to state transitions, they also note that “distributions of the unobserved random disturbances can be recovered by exploiting the condition that the observed choice of the agent maximizes utility”. We actually exploit the same idea but implement it with the objective of refining utility measures.

The choice made by the individuals to work j hours is observed. Since we assume that the individuals maximize their utility, we know that

$$v_{ij}(\cdot) > v_{ik}(\cdot) \quad \forall k \neq j, \quad (16)$$

$$u_{ij} + \epsilon_{ij} > u_{ik} + \epsilon_{ik} \quad \forall k \neq j. \quad (17)$$

The point is that we do not estimate $v_i(\cdot)$ but only its deterministic component u_i . By rearranging the terms, we find that

$$\epsilon_{ij} - \epsilon_{ik} > u_{ik} - u_{ij} \quad \forall k \neq j. \quad (18)$$

Two cases are possible. Suppose, for the *first case*, that the deterministic component of the utility of the individual i is not maximized at j but is dominated by some alternative choice k :⁵

$$\exists k \neq j : u_{ik} > u_{ij}. \quad (19)$$

In this situation, the second term of the inequality of Eq. 18 is positive, with the obvious consequence that $\epsilon_{ij} - \epsilon_{ik}$ has a positive lower bound. The interpretation is that, for an individual i who made the choice j , the stochastic component of his relative preferences $\epsilon_{ij} - \epsilon_{ik}$ should be large enough to compensate his relatively small deterministic group relative preference component (since $u_{ij} - u_{ik}$ is negative). As we detail below in this Section, this discrepancy between estimated preference and revealed preference provides an additional source of information that can be exploited to improve our estimate of the individual utility function.

For the *second case*, suppose on the contrary that the choice j made by the individual i does maximize the deterministic component of his utility:

$$u_{ij} > u_{ik} \quad \forall k \neq j. \quad (20)$$

In this situation, the second term of the inequality of Eq. 18 is negative, with the consequence that $\epsilon_{ij} - \epsilon_{ik}$ has now a negative lower bound. The interpretation is that the choice of the option j by the individual i is largely driven by his deterministic group relative preference component (since $u_{ij} - u_{ik}$ is positive $\forall k \neq j$), but also that the individual component ϵ_{ij} cannot be too small compared to any ϵ_{ik} . We also use this information to refine our measure of the individual preferences.

⁵ Note that the deterministic component of utility u_{ij} can be dominated in some cases by more than one alternative.

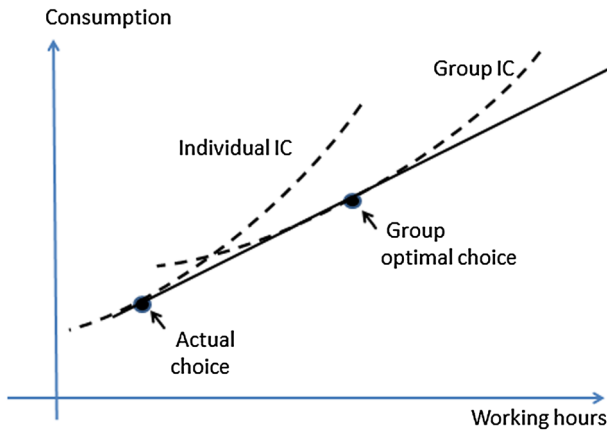


Fig. 6 Illustration of a discrepancy between group preference and revealed individual preference

We thus use the information derived from these two cases to improve our estimate of the individual utility function v_i . We therefore rely on the conditional logit assumption that ϵ_i is type 1 extreme value distributed to compute numerically⁶

$$E(\epsilon_{ij} - \epsilon_{ik} | \epsilon_{ij} - \epsilon_{ik} > u_{ik} - u_{ij}) \quad \forall i, k. \quad (21)$$

We can now use this information to improve the estimate of the utility function as detailed in Eq. 15, that we reproduce here for convenience

$$(\tilde{c}_{ik}, l_{ik}) = \left(\left[c_{ij}^{\alpha_c} + \frac{\alpha_c}{\beta_c} \left(\epsilon_{ij} - \epsilon_{ik} + \frac{\beta_l(z_i)}{\alpha_l} ((1 - l_{ij})^{\alpha_l} - (1 - l_{ik})^{\alpha_l}) \right) \right]^{\frac{1}{\alpha_c}}, l_{ik} \right), \quad (22)$$

where $\epsilon_{ij} - \epsilon_{ik}$ is no longer set to zero, but replaced by our estimate of $E(\epsilon_{ij} - \epsilon_{ik} | \epsilon_{ij} - \epsilon_{ik} > u_{ik} - u_{ij})$.

This procedure allows to improve the estimate of the utility function, necessary to compute the welfare measures, in particular the egalitarian equivalence criteria which require an estimate of the heterogeneous preferences. By extracting information from the potential discrepancy between the revealed choice and the optimal choice determined by the deterministic component of the utility, we go beyond the socio-demographic group preferences and closer to the individual preferences.

To illustrate the point of considering the individual preferences, we show in Fig. 6 the case of an individual, who decided to work much less (j hours) than what the deterministic/group indifference curve (Group IC) could suggest (k hours). This reveals that

⁶ We generate one million random numbers from a type 1 extreme value distribution to get a first set of ϵ 's, repeat the procedure to obtain a second set of ϵ 's, then take the differences, discard the values smaller or equal to $u_{ik} - u_{ij}$ and finally compute the respective averages of the differences to estimate $E(\epsilon_{ij} - \epsilon_{ik} | \epsilon_{ij} - \epsilon_{ik} > u_{ik} - u_{ij})$.

Table 1 Descriptive statistics on socio-demographic characteristics

Characteristics	Share (%)	Working hours	Wage
Less than high school	8.4	42.9	21,205
Education: high school	34.7	40.5	26,187
Education: >high school	56.9	41.4	36,693
White	56.4	41.5	34,760
Black	37.9	40.8	27,133
Other	5.7	41.0	32,506
Women	46.3	39.3	30,202
Men	53.7	42.9	33,073
Sample	100	41.2	31,745

Share stands for the sample share of individuals in the given socio-demographic category; working hours stands for the average working hours per week; Wage stands for the average annual net wage (\$)

the stochastic component of his utility function associated to j hours, ϵ_{ij} , should be larger than the one associated to k hours, ϵ_{ik} . Taking this information into account (following Eq. 22), we can approximate a new, individual specific, indifference curve (Individual IC) that turns out to be much steeper and better at explaining the actual choice of the individual.

5 Data and results

5.1 Data

The empirical analysis is based on US data from the Cross National Equivalent File (CNEF) for 2005 that provides information on incomes in 2004. It includes detailed information on socio-demographic variables. To avoid the problems related to considering multi-person households, we focus on singles without children. In addition, we restrict our sample to people who work at least 20 h a week.⁷ We also limit our sample to individuals aged between 20 and 65 years old who are not self-employed, retired or fully engaged in education. We finally exclude individuals whose capital income have a value superior to 10 % of the labor income.⁸ This gives us a sample of 914 individuals. Table 1 provides some descriptive statistics on the sample, where we see a quite large variability of the annual net wage across socio-demographic characteristics.

Even though we restrict our sample to individuals who work at least 20 h, we estimate our model by letting to the individuals the freedom to choose among all possible alternatives of working hours (including less than 20 h). The gross wage rate

⁷ As we have poor information on transfers given to people who do not work, we restrict our analysis to individuals least affected by transfers.

⁸ Capital income would affect the level of the budget set and probably the decision of working or not. Since we have no disaggregation or information on the source of capital income, and since taxes depend on the sources of this income, we choose to exclude the individuals with a capital income superior to 10 % of labor income.

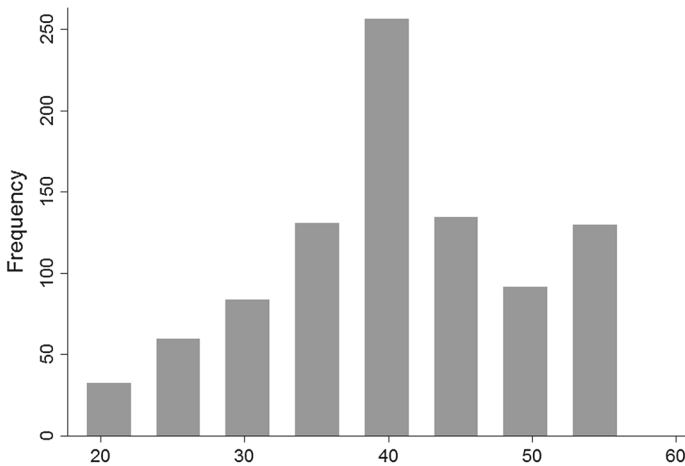


Fig. 7 Working hours

is assumed to be constant whatever the number of working hours. The distribution of working hours is illustrated in Fig. 7.

To build the budget set, we use the Taxing Wages 2005 OECD report to derive the net income for each possible working hours. More precisely, we first divide the individual gross labor income by the annual working hours to calculate the gross wage rate. We then discretize the labor time so as to reduce the choices of the individual to a finite number of discrete alternatives.⁹ For each discrete alternative, we compute the corresponding gross total income and we use the Taxing Wages 2005 OECD report to compute the corresponding net income.¹⁰ The discretized number of working hours is divided by 80, so that leisure is mapped in the space zero-to-one. Net income is expressed in units of 10,000 dollars per year.

We do not include capital income in the estimation. The gross amount of capital income the individual receives is available but there is no disaggregation by type of capital income and no data on net capital income. As taxes depend on the sources of this income, net capital income cannot be properly computed.

5.2 Estimation results

Table 2 presents the parameter estimates of the utility function for 6 variants. The benchmark case is estimated on the full sample, with heterogeneous preferences captured by the full set of socio-demographic variables (age, gender dummy, ethnic dummies and education dummies). We first find that the marginal utility of consumption and leisure ($\beta > 0$) is positive as expected, and decreasing ($\alpha < 1$), as expected.

⁹ The median of the empirical distribution in the following intervals define the discrete points: [17.5–22.5], [22.5–27.5], [27.5–32.5], [32.5–37.5], [37.5–42.5], [42.5–47.5], [47.5–52.5], > 52.5.

¹⁰ Seven tax brackets with rates of 0, 10, 15, 25, 28, 33 and 35% respectively, and bracket income limits at USD 8200, USD 15500, USD 29700, USD 71950, USD 150150 and USD 326450.

Table 2 Parameters of the utility function

Variables	(1)	(2)	(3)	(4)	(5)	(6)
Preference for consumption						
α_c	0.036 [0.030]	0.021 [0.038]	0.044 [0.038]	0.039 [0.030]	0.046 [0.029]	0.050* [0.029]
β_c	7.611*** [1.707]	7.571*** [1.695]	7.814*** [1.818]	7.574*** [1.703]	7.543*** [1.697]	7.520*** [1.704]
Preference for leisure						
α_l	0.466 [0.417]	0.190 [0.477]	0.260 [0.468]	0.458 [0.419]	0.459 [0.417]	0.455 [0.420]
Constant	17.754** [8.191]	13.814* [7.152]	14.839* [7.651]	17.518** [8.132]	17.613** [8.128]	17.459** [8.123]
Age	-0.511** [0.211]	-0.393** [0.187]	-0.406** [0.192]	-0.496** [0.207]	-0.494** [0.206]	-0.479** [0.202]
Age ²	0.006** [0.003]	0.005** [0.002]	0.005** [0.002]	0.006** [0.003]	0.006** [0.003]	0.006** [0.002]
Women	1.330*** [0.510]	1.449*** [0.559]	1.504*** [0.573]	1.335*** [0.509]	1.258** [0.493]	1.267*** [0.494]
Less than high school	0.274 [0.762]	0.096 [0.675]	0.051 [0.714]	0.326 [0.755]		
High school	0.754 [0.500]	0.650 [0.451]	0.674 [0.469]	0.758 [0.496]		
Black	0.318 [0.438]	0.129 [0.373]	0.120 [0.394]		0.339 [0.434]	
Other	0.229 [0.883]	-0.201 [0.770]	-0.391 [0.828]		0.115 [0.871]	
$\bar{\beta}_l$	9.213	7.355	8.193	9.142	9.169	9.151
Observations	914	853	848	914	914	914
Log likelihood	-1793.1	-1657.3	-1646.0	-1793.4	-1794.5	-1794.9

Standard errors in brackets.

Significant at 10%; ** significant at 5%; *** significant at 1%.

Specification (1) is the benchmark case. Workers earning less than 5 USD per hour are excluded from specification (2). Workers earning less than 5 USD per hour or more than 100 USD per hour are excluded from specification (3). Specifications (4), (5) and (6) are based on the benchmark case but vary on the regressors of β_l , the preference for leisure. $\bar{\beta}_l$ is the sample average of β_l . Less than high school and High school are dummy variables with More than high school as reference. Black and Other are dummy variables with White as reference value

We then note that the age and gender variables are the significant regressors capturing the heterogeneity of preferences for leisure. Being a woman, and being either old or young, is associated to a greater preference for leisure. High education has also a positive impact on the preference for leisure but is slightly insignificant. Other variables are clearly insignificant. These results are quite close to those reported in the [Decoster and Haan \(2015\)](#)'s study for Germany (except that they found that a low

education level has a significant positive impact on preference for leisure). Contrary to the results reported for Norway by [Aaberge and Colombino \(2014\)](#) in their recent survey, the marginal utility of leisure is found in our sample to be larger and less concave than the one of consumption. This comes not as a surprise given our specific sample of individuals working at least 20 h.

To the benchmark model, we add 5 variants to check the robustness of the estimates. The first variant (column 2) reports the estimates for a sample where individuals with hourly wages inferior to 5 USD are excluded. The second variant (column 3) excludes the same individuals and the ones earning more than 100 USD per hour. The variants reported in column 4, 5 and 6 rely on the benchmark full sample but exclude ethnic variables, education variables and both, respectively. No significant changes occur in the estimates. The same regressors are significant. Signs remain unchanged. On basis of the significance of the regressors, we keep the model presented in column 6 as our preferred model. The following steps rely on the estimates of this variant.

Last, we check the performance of the model by comparing the working hours maximizing the utility of the individuals with the actual choice of the individuals. We find that 57 % of the maximizing number of hours are equal or adjacent to the actual number of working hours. This relatively low percentage means that some unobserved preferences had an impact on individuals' choices.¹¹

5.3 Who are the worst-off?

The objective of this section is to identify the 91 individuals (10 % of the sample) to be considered as worst-off and to see how the reference values (for preferences in the conditional equality criterion and for the budget sets in the egalitarian equivalence criterion) affect this identification. We first report our results for the conditional equality criterion, then for the egalitarian equivalence criterion by emphasizing how accounting for individual preferences affects the results and finally make a comparison of the criteria.

The conditional equality criterion is implemented by fixing different reference values for β_l . We first use the sample first decile, average and last decile of β_l . We find that the identification of the worst-off based on these values is remarkably stable, with only minor changes (only 2 individuals out of 91 are affected). This is an interesting result for policy makers since the identification (in our sample) is relatively neutral to the choice of the reference value, at least if we take reasonable values for β_l .

We then compare our above results (where average $\beta_l = 10$) with more extreme values by increasing substantially the value of β_l and setting it to 20 and 70 successively, increasing thereby the preference for leisure. The results of this sensitivity analysis are reported in Fig. 8 where we see that 782 individuals of our sample are never considered as worst-off, that 50 individuals only are commonly considered as

¹¹ A part of this low percentage is due to individual heterogeneous preferences, but another part is also due to the imperfect fitting of the model to the data. It can come from different sources including (i) the Cox-Box specification of the deterministic part of the utility function, (ii) the determination of the budget constraint (including the neglect of capital income), and (iii) the discretization of labor supply. There is unfortunately no convincing way of disentangling the respective contribution of these factors.

Fig. 8 Conditional equality—impact of β_l on the identification of the worst-off

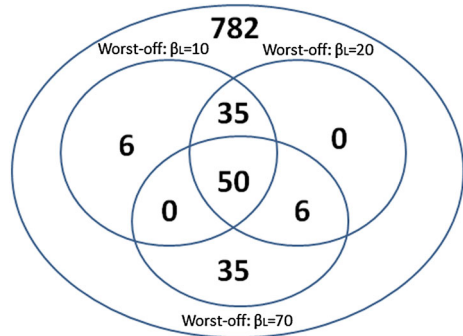


Table 3 Conditional equality—wage and working hours of the worst-off

Worst-off	Average wage	Average working hours
CE 10	4.41	41.95
CE 20	4.57	40.70
CE 70	8.13	34.85

CE 10, CE 20 and CE 70 stand for conditional equality criteria with preference for leisure determined by the reference values β_l equal to 10, 20 and 70, respectively. Columns 2 and 3 report the average wage per hour (\$) and working hours per week of the individuals identified as worst-off by the CE 10, CE 20 and CE 70 criteria

worst-off by the three criteria and that 41 individuals are considered as worst-off by the low preference for leisure reference values ($\beta_l = 10$ and $\beta_l = 20$) and 41 others by the (very) high preference for leisure reference value ($\beta_l = 70$).

The choice of the reference value determines who will be categorized as worst-off. As detailed in Sect. 2, the lower (higher) the reference preference for leisure, the more likely the low(high)-wage individuals will be considered as worst-off. This is confirmed in Table 3 where we see that the average wage of the worst-off increases with the reference value of β_l , the parameter capturing the preference for leisure.

We finally check whether the reference value has an impact on the socio-demographic profile of the individuals identified as worst-off (gender, age, education). We find that the profiles remain very stable even for extreme values for β_l .¹²

We then implement the egalitarian equivalence criterion and compare it “with” and “without” consideration of the individual preferences. Table 4 shows the under-representation (<10%) or over-representation (>10%) of the worst-off per social category for the zero-wage, the minimum wage and the wage egalitarian equivalence criteria.

We find that the choice of the egalitarian equivalence criterion is not neutral. As an example, by comparing the zero-wage and minimum wage cases (where the minimum wage rate is set to 5 dollars per hour), we see that the worst-off tend to be over-represented in the women category, according to the former criterion, and under-

¹² Results not reported, available on request.

Table 4 Egalitarian equivalence—comparison of the socio-demographic profiles of the worst-off with and without consideration of the unobserved heterogeneity

	Zero EE		Min EE		Wage EE	
	With (%)	Without (%)	With (%)	Without (%)	With (%)	Without (%)
Women	11.6	11.6	9.2	9.0	9.5	9.9
Men	8.6	8.6	10.6	10.8	10.4	10.0
Black	13.3	13.3	14.7%	14.5	14.7	14.2
White	8.1	7.8	6.8	7.0	6.8	7.2
Other	5.8	9.6	9.6	9.6	9.6	9.6
Less than high school	20.8	19.5	20.8	20.8	20.8	19.5
High school	12.9	14.2	13.9	13.9	13.9	13.6
More than high school	6.5	6.0	6.0	6.0	6.0	6.3

Zero EE, Min EE and Wage EE stand for zero-wage, minimum wage and wage egalitarian equivalence criteria, respectively. The “With” and “Without” headers refer to the approaches where the unobserved heterogeneity is considered, respectively not considered. A percentage below (above) 10 % indicates that the worst-off are under-represented (over-represented) in that socio-demographic category

represented according to the later. This implicitly shows that the women in our sample have a higher distaste for working, since lower reference wages tend to categorize as worst-off those with a low preference for working.

We also note that considering the individual preferences (“With” columns) instead of group preferences only (“Without” columns) seems to have no substantial impact on the socio-demographic profile (gender, race, education) of those identified as worst-off for any of the three criteria. This apparent stability of the profiles categorized as worst-off does not guarantee a stability of the individuals themselves. Indeed, we show in Fig. 9 that this apparent stability actually hides a relative variability for the zero-wage egalitarian equivalence where 17 individuals (18.7 % of the worst-off) are no longer considered as worst-off once we consider the individual preferences in the estimation. We also note that the individual variability is smaller for minimum wage and wage egalitarian equivalence criteria.

We now check the consistency of these criteria in the identification of the worst-off in Fig. 10 and then compare them to conditional equality criteria in Fig. 11.

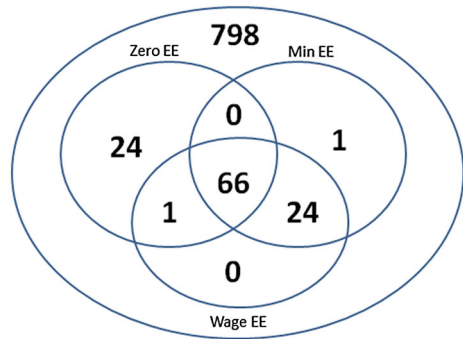
We find that 798 individuals are never categorized as worst-off by any of the criteria, that zero-wage egalitarian equivalence is quite specific, with 24 individuals not identified as worst-off by the other two criteria. We also note that minimum wage and wage egalitarian equivalence criteria identify the same individuals as worst-off (with one exception).

To get a better grasp of what lies behind this specificity of the zero-wage compared to other egalitarian equivalence criteria, we report in Table 5 the average wage and working hours of the individuals categorized as worst-off by the egalitarian equivalence criteria (with consideration of individual preferences). We find again that the worst-off identified by the minimum wage and wage egalitarian equivalence criteria are very similar and that the average wage of the worst-off identified by the zero-wage egalitarian equivalence criterion is slightly larger than the one of the other criteria. As



Fig. 9 Egalitarian equivalence—impact of considering (“With”), not considering (“Without”), the unobserved heterogeneity on the identification of the worst-off

Fig. 10 Egalitarian equivalence—consistency in the identification of the worst-off



expected we find that lower reference wages tend to categorize as worst-off those with a low preference for working (41.1 h for zero-wage and 42.4 for minimum wage).

Regarding the comparison exercise, reported in Fig. 11, between egalitarian equivalence and conditional equality criteria, the main results are the following. The conditional equality criterion with average reference value (*CE10*) and the minimum wage and wage egalitarian equivalence criteria (*EE_m* and *EE_w*) are very consistent in the identification of the worst-off (only 4 exceptions). The conditional equality criterion with (very) high preference for leisure reference value (*CE70*) identifies very specific individuals not captured by the egalitarian equivalence criteria. The zero-wage egalitarian equivalence criterion (*EE0*) is not fully consistent with the conditional equality criteria.

To summarize, we note the following: (1) taking into account individual preferences can substantially affect the identification of the worst-off - this was particularly the case

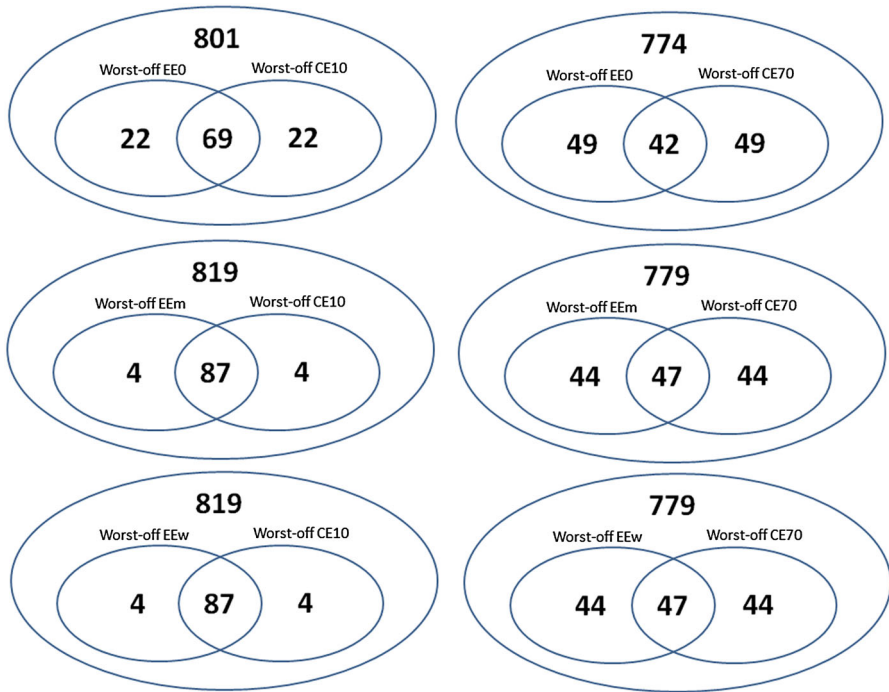


Fig. 11 Egalitarian equivalence and conditional equality—consistency in the identification of the worst-off

Table 5 Egalitarian equivalence—wage and working hours of the worst-off

Worst-off	Average wage	Average working hours
Zero EE	4.83	41.11
Min EE	4.38	42.41
Wage EE	4.37	42.35

Zero EE, Min EE and Wage EE stand for zero-wage, minimum wage and wage egalitarian equivalence criteria, respectively. Columns 2 and 3 report the average wage per hour (\$) and working hours per week of the individuals identified as worst-off by the Zero EE, Min EE and Wage EE criteria

for the zero-wage egalitarian equivalence criterion; (2) conditional equality criteria are relatively consistent in the identification of the worst-off as long as the preference for leisure reference values are reasonable (i.e. drawn from the sample), but consistency vanishes with more extreme reference values; (3) minimum wage and wage egalitarian equivalence criteria are relatively similar in the identification of the worst-off but differ from the zero-wage egalitarian equivalence criterion.

6 Conclusions

This paper presents an empirical exercise on the conditional equality and egalitarian equivalence criteria. We propose an extension of the empirical methodology developed

by Bargain et al. (2013) and Decoster and Haan (2015) to implement the egalitarian equivalence criterion of Fleurbaey (2008). The limitation of their approach was to consider only socio-demographic group preferences and not individual ones. We show that part of the individual preferences can be retrieved and incorporated into the analysis. Information on the individual preferences is derived from the potential discrepancy between the group level optimal choice and the revealed individuals' choice. We also implement the conditional equivalence criterion and compare our results.

We apply our model to a sample of singles from the 2004 US CNEF dataset and detail the profiles of the worst-off. We show that the identification of the worst-off by conditional equality and egalitarian equivalence criteria is quite consistent (with the relative exceptions of the zero-wage egalitarian equivalence criterion and the conditional equivalence criterion with (very) large preference for leisure). We more specifically show for the egalitarian equivalence criterion that going beyond socio-demographic group preferences and better estimating individual preferences affects the identification of the worst-off. As an illustration, we find that 18.7 % of the worst-off identified by the zero-wage egalitarian equivalence drop from the worst-off category when we account for individual-specific preferences.

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